

Bayesian Inference

INFO 521 · Module 4 · Lecture B — The Conjugate Gaussian
Posterior

Greg Chism

College of Information Science · University of Arizona

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Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

Learning outcomes

By the end of Lecture B you should be able to:

1. **Place a Gaussian prior on the weights** and state the conjugate posterior $\mathcal{N}(\mathbf{w} \mid \mathbf{m}_N, \mathbf{S}_N)$.
2. **Read the update formulas** — what shrinks, what dominates, and why $\mathbf{X}^\top \mathbf{X}$ appears *again*.
3. **Show ridge regression is MAP** with a Gaussian prior ($\lambda = \sigma^2 / \tau^2$).
4. **Form the posterior predictive** and separate its two variance terms.

From θ to w

Lecture A put a distribution on a *proportion*. Now put one on the **regression weights** of the Module-1 model:

$$\text{prior: } p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{m}_0, \mathbf{S}_0)$$

$$\text{likelihood: } p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(y_n \mid \mathbf{w}^\top \mathbf{x}_n, \sigma^2)$$

- $\mathbf{m}_0$ — where belief starts (e.g., “SBP near 120, no age effect”).
- $\mathbf{S}_0$ — how *loosely* it starts (wide diagonal = weakly informative).
- σ^2 — treated as **known** today (use m3a’s $\hat{\sigma}^2$); it keeps the algebra conjugate.

The conjugate Gaussian posterior

Gaussian prior $\times$ Gaussian likelihood $\rightarrow$ **Gaussian posterior**, in closed form:

$$p(\mathbf{w} \mid \mathcal{D}) = \mathcal{N}(\mathbf{w} \mid \mathbf{m}_N, \mathbf{S}_N)$$

$$\begin{aligned}\mathbf{S}_N &= (\mathbf{S}_0^{-1} + \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X})^{-1} \\ \mathbf{m}_N &= \mathbf{S}_N (\mathbf{S}_0^{-1} \mathbf{m}_0 + \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{y})\end{aligned}$$

Project 1 ends exactly here. The full pipeline — data $\rightarrow$ model $\rightarrow$ loss $\rightarrow$ likelihood $\rightarrow$ posterior — closes on this pair of formulas.

Reading the formulas

Work in **precisions** (inverse covariances) and the update is addition again:

- $\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X}$ — **precisions add**: prior information + data information.
- The data's contribution $\frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X}$ is the **Fisher information** $\mathcal{I}$ from m3b — the same object, fourth appearance.
- $\mathbf{m}_N$ is a **precision-weighted blend** of $\mathbf{m}_0$ and the data — the vector version of m4a's shrinkage identity.
- $N \rightarrow \infty$: the data term dominates, $\mathbf{m}_N \rightarrow \hat{\mathbf{w}}_{\text{MLE}}$, $\mathbf{S}_N \rightarrow \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} = \text{Cov}[\hat{\mathbf{w}}]$.

Watching the posterior contract

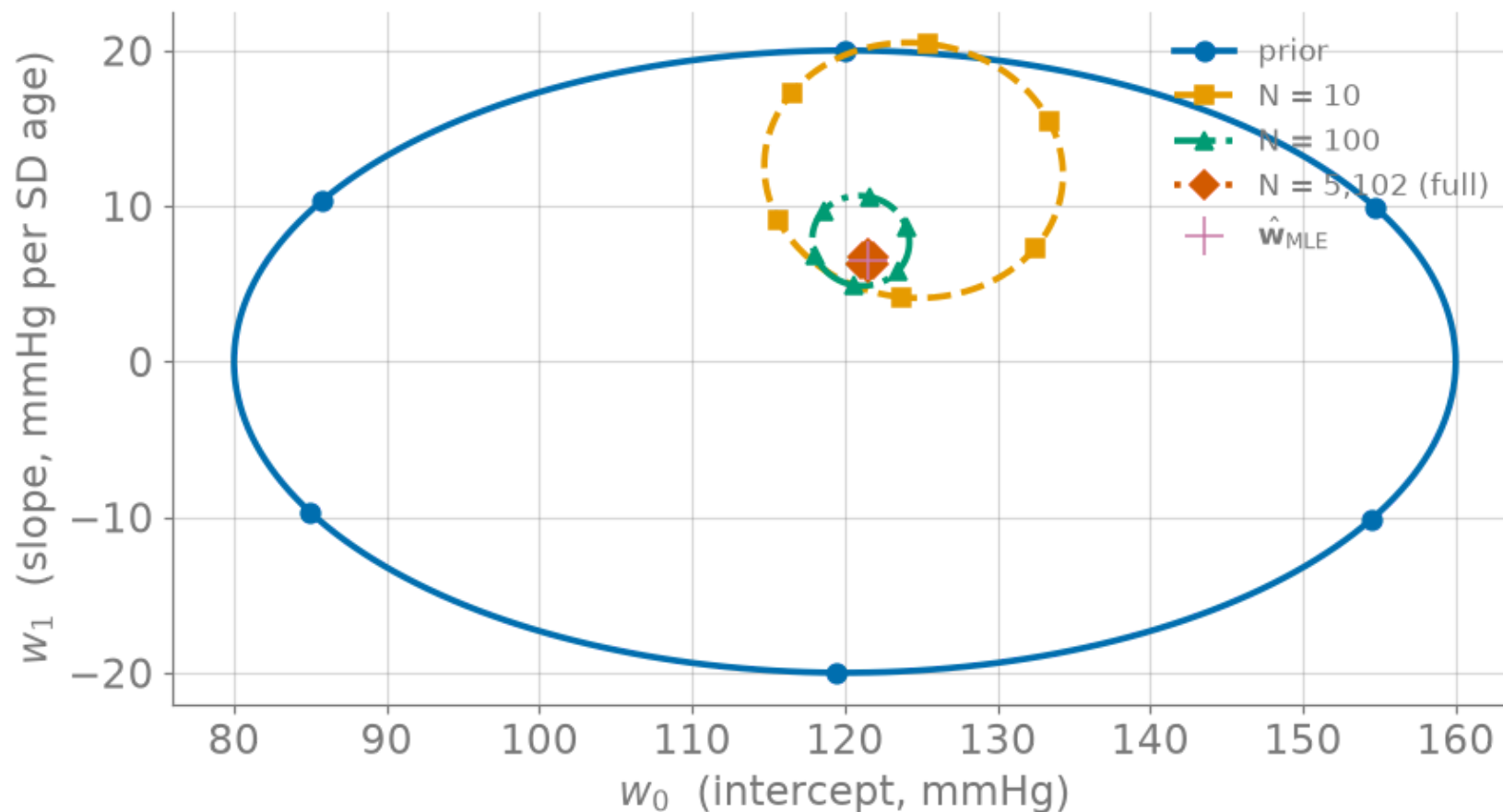


Figure 1: Conjugate posterior over (w_0, w_1) , real NHANES draws (seed 521): prior $\rightarrow$ $N = 10 \rightarrow 100 \rightarrow 5,102$. The purple + is the MLE.

Special case: the ridge prior

Take a **zero-mean isotropic** prior, $\mathbf{m}_0 = \mathbf{0}$, $\mathbf{S}_0 = \tau^2 \mathbf{I}$:

$$\mathbf{m}_N = \left(\mathbf{X}^\top \mathbf{X} + \frac{\sigma^2}{\tau^2} \mathbf{I} \right)^{-1} \mathbf{X}^\top \mathbf{y} = \hat{\mathbf{w}}_{\text{ridge}} \quad \text{with} \quad \boxed{\lambda = \frac{\sigma^2}{\tau^2}}$$

- m2b's forward hook, **cached in**: the ridge penalty *is* a Gaussian prior.
- λ stops being a dial and becomes a *statement*: noisy data (big σ^2) or a tight prior (small τ^2) $\Rightarrow$ shrink harder.

MAP vs. the full posterior

The **MAP** (maximum a posteriori) estimate is the posterior's peak:

$$\hat{\mathbf{w}}_{\text{MAP}} = \arg \max_{\mathbf{w}} [\ell(\mathbf{w}) + \log p(\mathbf{w})] = \mathbf{m}_N \quad (\text{Gaussian case: peak} = \text{mean}).$$

- **MAP is one number** — regularized fitting in Bayesian clothes; it discards the spread.
- **The posterior is the answer** — $\mathbf{S}_N$ carries what MAP throws away, and predictions (next slide) *use* it.
- Gaussians hide the difference (peak = mean); Module 5's posteriors won't be Gaussian, and the gap becomes the whole story.

Predicting a new patient

For a new input $\mathbf{x}_*$, integrate the model over the posterior:

$$p(y_* \mid \mathbf{x}_*, \mathcal{D}) = \mathcal{N} \left(y_* \mid \mathbf{m}_N^\top \mathbf{x}_*, \underbrace{\sigma^2}_{\text{noise}} + \underbrace{\mathbf{x}_*^\top \mathbf{S}_N \mathbf{x}_*}_{\text{parameter uncertainty}} \right)$$

Two variance terms, two different fixes:

- σ^2 — patient-level scatter; **irreducible** (m2b's floor).
- $\mathbf{x}_*^\top \mathbf{S}_N \mathbf{x}_*$ — *our* ignorance of $\mathbf{w}$; **shrinks with data**, grows away from the data's center.

The predictive band on NHANES

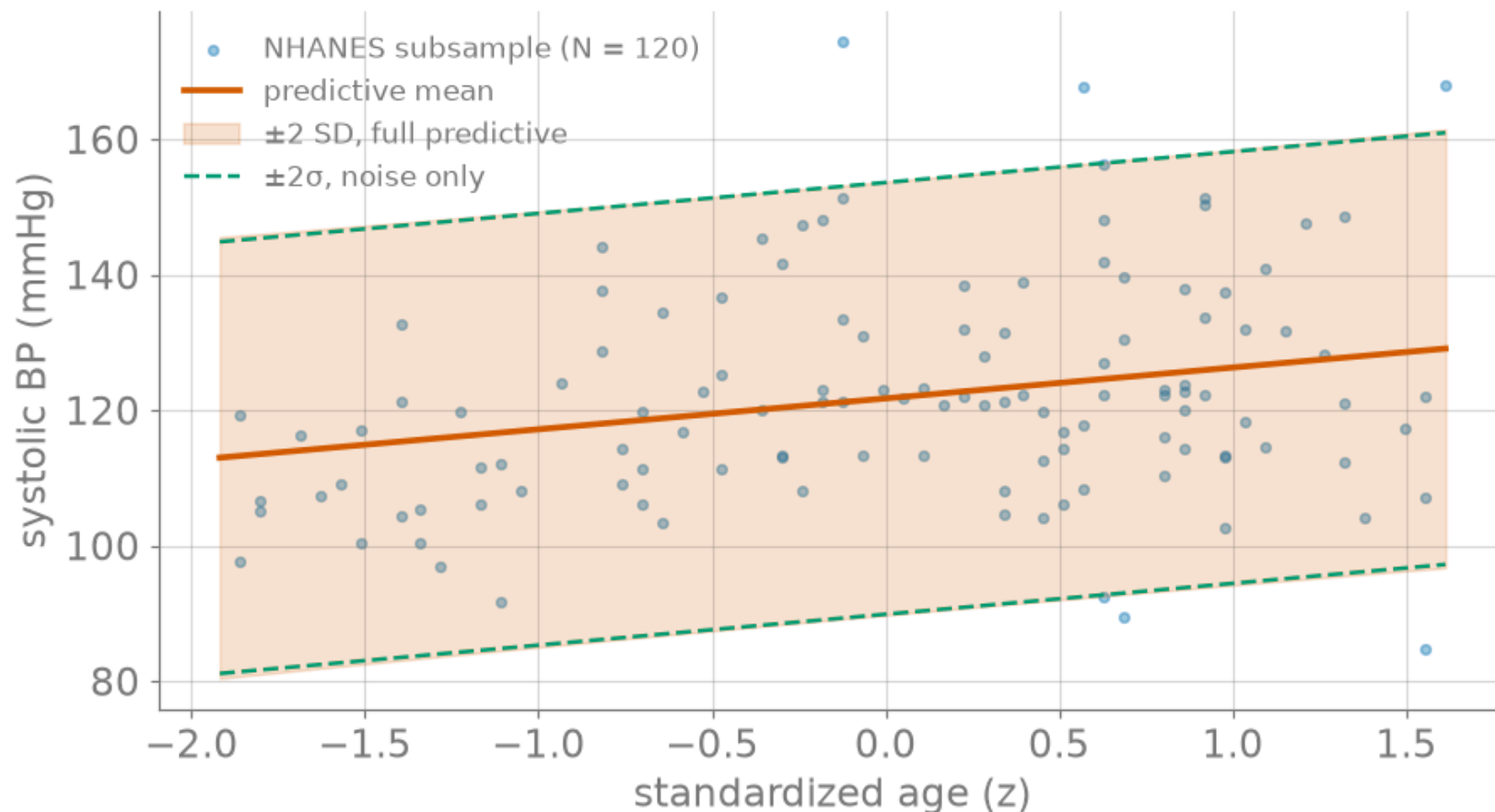


Figure 2: Posterior predictive from N = 120 real patients: mean ± 2 predictive SD (shaded) vs. noise-only $\pm 2\sigma$ (green dashes) — parameter uncertainty flares at the extremes.

The evidence, briefly

m4a's ignored denominator, $p(\mathcal{D})$ — the **marginal likelihood** or *evidence* — is the likelihood **averaged over the prior**:

$$p(\mathcal{D}) = \int p(\mathcal{D} \mid \mathbf{w}) p(\mathbf{w}) d\mathbf{w}$$

- Scores a **whole model** (prior included), not one parameter value.
- Comparing evidences ranks models **without a test set** — an automatic Occam's razor: too-flexible models spread their prior thin and score lower.
- For today's conjugate model it, too, is closed-form; beyond, it is the first casualty (Module 5).

Recap & bridge

- **Conjugate Gaussian posterior:** $\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X}$, $\mathbf{m}_N = \mathbf{S}_N(\mathbf{S}_0^{-1} \mathbf{m}_0 + \frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{y})$ — Project 1's endpoint.
- **Ridge = MAP** with $\mathbf{S}_0 = \tau^2 \mathbf{I}$: $\lambda = \sigma^2 / \tau^2$.
- **Predictive variance** $= \sigma^2 + \mathbf{x}_*^\top \mathbf{S}_N \mathbf{x}_*$ — noise + ignorance, separately visible.

Next (Module 5): binarize the outcome — hypertensive or not — and the Gaussian likelihood is gone. Conjugacy **breaks**, no closed form survives, and approximation becomes the craft.